Language Model Supported Data Scraping Tool for Wastewater Related Contaminants

Introduction: Background

Managing the risks of de facto wastewater reuse nationally remains a significant challenge due to the need for systematic methods and routine monitoring of many biological and chemical contaminants discharged with wastewater. These risks are further amplified under extreme climate conditions, such as severe droughts that reduce natural flow and limit the dilution of contaminants and flooding events that trigger combined sewer overflows, worsening contamination.

To address this, we propose a new approach that will harness the power of language models to scrape and compile data on wastewater contaminants from the vast body of existing literature. This data can then serve as the foundation for training Machine Learning models capable of predicting wastewater contaminant loading for future use. By bridging the critical data gaps, this predictive framework would offer a powerful tool to evaluate the contamination risks more effectively, particularly under the pressures of changing environmental conditions.

Data Collection/Pre-Processing

crossref_results_w_link.csv

Title object	DOI object	Publication Date o.	Author object	URL object	Publisher object
A year-long study...	10.1016/j.scitoten...	2023	Helena Rapp-Wri...	https://doi.org/1...	Elsevier BV
Antibiotics in Was...	10.3390/toxics10...	2022	Bondi Gevao; Saif...	https://doi.org/1...	MDPI AG
Concentration of ...	10.17504/protocol...	Not available	Katy Graham; Ma...	https://doi.org/1...	ZappyLab, Inc.
Metal-Organic Fr...	10.1002/9783527...	2023	Khushbu Sharma;...	https://doi.org/1...	Wiley
Emerging Conta...	10.1007/978-3-0...	2022	Revathy Rajakum...	https://doi.org/1...	Springer Internati...
Effects of Sewag...	10.2139/ssrn.465...	Not available	Seyed Mohsen K...	https://doi.org/1...	Elsevier BV
Application of wa...	10.5004/dwt.202...	2021	Madani Bessedik;...	https://doi.org/1...	Elsevier BV
Modeling the infl...	10.1007/s00500-...	2023	Behrouz Abolpou...	https://doi.org/1...	Springer Science ...
A survey on the r...	10.1080/19443399...	2015	Ali Almasi; Kiomar...	https://doi.org/1...	Elsevier BV
Pharmaceutical ...	10.3390/toxics50...	2017	Okhumode Yakubu	https://doi.org/1...	MDPI AG

- Collected academic literature and metadata from Springer and CrossRef using automated tools.
- Scraped bibliographic details and full-text data on wastewater reuse and contaminants.
- Utilized LangChain and PyPDFLoader to extract text and metadata from unstructured PDF documents, converting them into machine-readable formats
- Used Scrapy, Selenium, and URL manipulation techniques to extract online literature, storing them as PDF documents systematically.
- Implemented an LLM wrapper to facilitate advanced search capabilities, streamlining the collection and extraction of relevant literature.

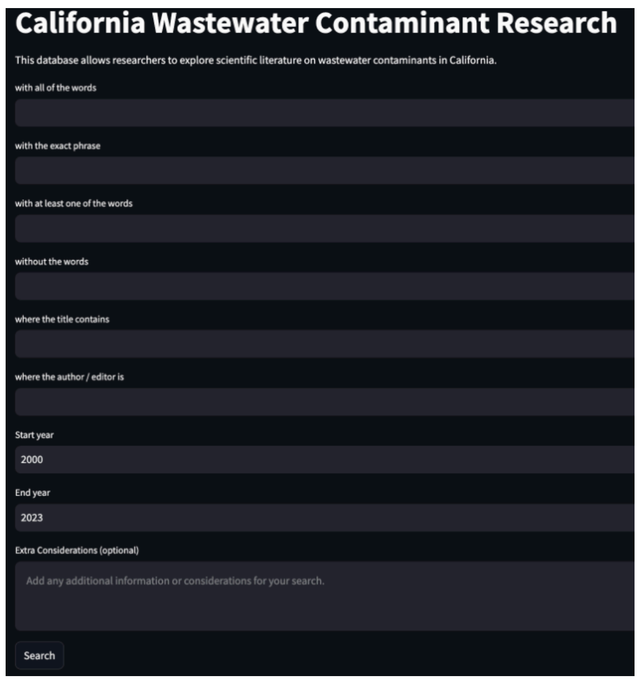
Download Automation

- The automated information extraction process utilizes Selenium (IDE & WebDriver) & PyAutoGUI. It starts by identifying articles on wastewater contaminants, either by retrieving the top n articles from a publisher's website based on specific criteria or using a text file of article links.
- Selenium accesses each article's webpage to extract the full text, tables, and links to associated images or figures from the source code, generating separate output files for the text, tables, and image links of each paper.
- Selenium and PyAutoGUI process the image links by visiting each URL, capturing screenshots, and uploading the images to a Large Language Model (LLM) with a predefined prompt. The LLM's responses are saved alongside the corresponding image links.
- At the end of the workflow, each paper is processed to produce comprehensive outputs: the full text, extracted tables, and summaries of all images and figures, ensuring efficient and structured data extraction.
- A series of trials and errors with the libraries were used to extract the information from source code and generating summaries from the images.

Information Retrieval

- Leveraged AzureChatOpenAI to analyze the extracted content, using user-defined prompts to provide contextually relevant insights, structured datasets, and key information, ensuring that user queries were accurately addressed.
- Implemented a Retrieval-Augmented Generation (RAG) strategy, embedding document vectors into a vector database for similarity-based retrieval, and feeding the retrieved content to an LLM to generate detailed, structured responses, supporting efficient and precise information retrieval

User Interface



Integrated features like dynamic (filtering) search capabilities and real-time data visualization.

Increased awareness about wastewater reuse risks through intuitive and accessible data presentations.

Enabled stakeholders to interact with findings, fostering collaboration and improving usability.

Accessible interface to explore data insights, including contaminant distributions, metadata, and predictive modeling results.

Simplified complex data for users through a clean, user-friendly interface and streamlined navigation.

Storage

- Extracted document metadata, including text summaries, using an LLM wrapper. Generated and stored summary embeddings using a FAISS vector database, along with full text and metadata in a local CSV file for efficient access.
- Embedded user prompts and compared them with stored document summary embeddings to identify the most relevant documents, which were used as reference sources for LLM-generated responses.
- Used Google Firebase to store large-scale amounts of metadata and extracted information from articles in structured JSON format
- Fully integrated with scraping and processing tools for optimized data pipeline
- Implemented Firebase Authentication to safeguard data integrity and privacy

Conclusion: Future Improvements

- Provide dynamic visualizations (e.g., keyword trends, citation networks) to help users understand patterns and relationships within the retrieved data
- Fine-tune language model to better align with user's input requirements to capture nuanced aspects of wastewater contaminants
- Expand and integrate other external data sources to increase scope of received articles
- Integrate and leverage summarization models/libraries to generate concise summaries of extracted information
- Explore additional similarity search methods, such as keyword matching, and consider integrating knowledge graphs with LLM to improve output accuracy and reliability.

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